

CODE:

```
import numpy as np

# Code sequences
c1 = [1, 1, 1, 1]
c2 = [1, -1, 1, -1]
c3 = [1, 1, -1, -1]
c4 = [1, -1, -1, 1]

rc = []

print("Enter the data bits:")

d1 = int(input("Enter D1: "))
d2 = int(input("Enter D2: "))
d3 = int(input("Enter D3: "))
d4 = int(input("Enter D4: "))

# Encoding
r1 = np.multiply(c1, d1)
r2 = np.multiply(c2, d2)
r3 = np.multiply(c3, d3)
r4 = np.multiply(c4, d4)

resultant_channel = r1 + r2 + r3 + r4
print("Resultant Channel:", resultant_channel)

# Select channel
Channel = int(input("Enter the station to listen for C1=1, C2=2, C3=3, C4=4: "))

if Channel == 1:
    rc = c1
elif Channel == 2:
    rc = c2
elif Channel == 3:
    rc = c3
elif Channel == 4:
    rc = c4

# Decoding
inner_product = np.multiply(resultant_channel, rc)
print("Inner Product:", inner_product)

res = sum(inner_product)
data = res / len(inner_product)

print("Data bit that was sent:", data)
```

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Name	Modified
cpp-small.ipynb	2mo ago
cpp-third-party-li...	2mo ago
cpp.ipynb	2mo ago
exep1.ipynb	now
Intro.ipynb	2mo ago
Lorenz.ipynb	2mo ago
r.ipynb	2mo ago
sqlite.ipynb	2mo ago
Untitled1.ipynb	now

```
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c4 = [1, -1, -1, 1]
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# Encoding
r1 = np.multiply(c1, d1)
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resultant_channel = r1 + r2 + r3 + r4
print("Resultant Channel:", resultant_channel)

# Select channel
Channel = int(input("Enter the station to listen for C1=1, C2=2, C3=3, C4=4: "))

if Channel == 1:
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# Decoding
inner_product = np.multiply(resultant_channel, rc)
print("Inner Product:", inner_product)

res = sum(inner_product)
data = res / len(inner_product)

print("Data bit that was sent:", data)

Enter the data bits:
Enter D1: 1
Enter D2: 1
Enter D3: 1
Enter D4: 1
Resultant Channel: [4 0 0 0]
Enter the station to listen for C1=1, C2=2, C3=3, C4=4: 1
Inner Product: [4 0 0 0]
Data bit that was sent: 1.0
```